Design Brief – Review 2

0. Header

Project code	FA26SE032
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Repository link	https://github.com/thang-nt25/SCANMS-project-capstone-fall26 (tag: review-2)
Version	2.0
Date	06/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Timeline & Team (Sec. 1–2)	09/2026–03/2027; 5th unassigned row	07/09/2026–30/12/2026; finalized 4 members	Match Fall 2026 defense calendar and lock team roster.
Domain Model & Pricing (Sec. 3.2a–b)	Open multi-seller portal; unguarded discounts	4 in-house brand divisions, Category Lock, EAN-13 check & MAP floor-price promotions	Focus strictly on enterprise collaborator management, not an open C2C marketplace.
5-Role IAM & SoD (Sec. 3.2b–c)	3 roles; public shop signup; guest checkout	5 roles; Admin-only staff provisioning; KOC eKYC (Bronze 10%); 0-loss auth checkout	Prevent external shop creation and ensure verified tax/delivery identity.
Remuneration & Attribution (Sec. 3.2b)	Simple/additive rates; generic vouchers	Best-Benefit Precedence (cap 25%), KOC coupon [KOC][5|10], Self-Referral = 0%, KPI escrow	Protect brand margins, ensure cookie-less attribution, and block self-buying.
Returns, Escrow & Payout (Sec. 3.2b–c)	Linear payout; no return split; manual file only	7-day hold; 1-1 Exchange (keeps KOC fee) vs Refund (clawback); 10% PIT & BIDV Sandbox	Fairly protect KOCs on carrier damage while preventing overpayment on refunds.
Realtime, AI, Cloud & Mobile (Sec. 3.2b,f)	Static web portal; no chat/SLA/AI/mobile	Socket.IO VIP deals (>16% Admin gate), 14d Sample SLA, Dual AI, Cloud Web + Expo Mobile App	Support real-time partner workflows across deployed Web and Android clients.

1.2 Out of Scope

Item	Reason
Open External Seller Marketplace & Public Shop Registration	SCANMS governs external collaborators for in-house enterprise brand divisions; external third-party merchants cannot open public stores.
Generic Retail Voucher Stacking & Standalone Marketing Portals	Platform/freeship voucher stacking and standalone promotional portals are excluded to prevent margin cannibalization; only MAP-guarded ProductPricePromotion and KOC-bound attribution coupons are supported.
Factory Manufacturing, Supplier Procurement (ERP/AP) & Robotics WMS	Upstream OEM manufacturing, purchase orders, and warehouse hardware belong to corporate ERP; SCANMS manages sellable stock quotas and SKU floor prices.
Proprietary Courier Vehicles & Licensed Core Banking Rails	Physical parcel delivery and fiat banking rails are delegated to GHN Sandbox API, PayOS (order collection), and BIDV Sandbox API (collaborator payouts).

2. System at a Glance

SCANMS solves fragmented partner onboarding, attribution fraud, margin erosion, and manual commission settlement for multi-brand D2C retail enterprises managing external sales collaborators (KOLs/KOCs). Built for five roles—System Administrator, System Manager, In-House Shop Manager, Collaborator, and Customer—across deployed Web portals (React/Vite) and a native Android application (React Native/Expo), SCANMS automates the entire affiliate lifecycle across four company-owned brand divisions. Core capabilities include Admin-governed staff provisioning, Maker-Checker eKYC, category-locked catalogs, MAP floor-price promotions, 4-axis AI partner matchmaking, Socket.IO VIP deal negotiation with >16% Admin budget escalation, 14-day sample review SLAs, Smart Link/QR and KOC-coupon attribution, 7-day post-delivery commission holds with 1-for-1 exchange vs. refund clawback, and row-locked payouts with 10% PIT withholding via BIDV Sandbox. See System Overview Architecture Diagram in SDD (Report 4, Section 3.1).

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	In-House Brand Provisioning & Shared Mobile/Web Bootstrap	Public seller self-signup & separate mobile BFF vs Admin-only staff provisioning & unified NestJS API	Blocked public SHOP_MANAGER signup (/admin/personnel only) and exposed /api/mobile/bootstrap over one NestJS modular monolith so Web and Expo Android enforce identical 5-role RBAC.	Nguyễn Thành Thắng	F-01, F-02 / SDD Sec. 3.1
D2	MAP Floor-Price Guard & KOC-Exclusive Attribution Coupon	Unrestricted shop price edits & multi-voucher stacking vs Maker-Checker MAP guard & KOC-bound coupon	Enforced proposedPrice >= floorPrice with System Manager approval (ProductPricePromotion) and single KOC goods coupon ([KOC][5|10]) for deterministic cookie-less attribution.	Nguyễn Thành Thắng	F-06, F-13 / SDD Sec. 4.2
D3	Socket.IO VIP Deal Negotiation with >16% Admin Escalation	HTTP polling & unguarded deal rates vs WebSocket rich deal cards with 2-tier approval gate	Used Socket.IO /chat rooms with autonomous activation for 10%–16% deals and mandatory PENDING_ADMIN_APPROVAL escalation to System Admin for 17%–25% deals.	Phan Xuân Thịnh	F-09, F-10 / SDD Sec. 5.3
D4	7-Day Post-Delivery Hold & Dual-Path Return Resolution	Instant commission release or blanket clawback on all claims vs 7-day hold with Exchange vs Refund split	Held pending commissions 7 days post-delivery: 1-for-1 EXCHANGE on carrier damage preserves 100% KOC commission (+ GHN claim), whereas REFUND triggers automatic clawback.	Nguyễn Phú Quý	F-16, F-17 / SDD Sec. 5.4
D5	Best-Benefit Commission Precedence & Row-Locked BIDV Payout	Additive rate stacking & unlocked balance updates vs Precedence override, KPI escrow & SELECT FOR UPDATE	Applied min(25%, max(VIP Deal, 10%+Tier, 10%) + Live), isolated VIP bonus in conditionalDealEscrowBalance, and locked wallet rows with 10% flat PIT withholding via BIDV Sandbox.	Nguyễn Đình Tuấn	F-17, F-19 / SDD Sec. 6.2

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Self-Referral Purchases & Bot Click Flooding distorting attribution and draining marketing budgets.	High: Unfair commission leakage and skewed AI conversion rates (CR%).	isSelfReferral guard forces self-purchase commission to 0%; Redis sliding-window limit (100 req/min/IP) with RAM LRU fallback and AiFraudService anomaly flags.	Nguyễn Phú Quý
R2	Brand Margin Erosion from below-cost discounts or excessive VIP collaborator rates.	High: Negative unit economics on discounted SKUs.	assertFloor(proposedPrice, floorPrice) Maker-Checker approval, Best-Benefit non-additive cap (<=25%), >16% Admin approval gate, and KPI-locked deal bonus escrow.	Nguyễn Thành Thắng
R3	Concurrent Withdrawal Double-Spending, Unreturned Samples, or Refund Overpayment.	High: Wallet overdrafts, sample inventory loss, and unrecovered commissions.	PostgreSQL SELECT ... FOR UPDATE wallet locks, idempotent BIDV Sandbox inquiry, 7-day return hold clawback, and 14-day sample SLA overdue quota lock.	Nguyễn Đình Tuấn

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	SDD ERD & Architecture Docs (21-table draft) lag behind 69-model Prisma Schema & Expo Mobile client (BM02 SR-01, SR-02)	docs/architecture/SAD_DOCUMENT.md, DATABASE_DESIGN.md vs schema.prisma & Mobile/	Regenerate Logical ERD, State Machines, and System Overview diagram in Report 4 to match 69 Prisma models, Vite Web, Expo Mobile, and @nestjs/schedule.	Nguyễn Thành Thắng	Defense 1.1
G2	Mobile Android App defers Customer PayOS Checkout & Live Room to Web Portal in current iteration (BM02 SR-04)	Mobile/src/features/cart/CartScreen.tsx, Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md	5-role Mobile operations (catalog, KYC, chat, VIP deals, samples, pricing, GHN orders, returns, payouts) are active; complete native checkout deep-link handoff & staging APK build.	Phan Xuân Thịnh	Defense 1.1
G3	Commercial Host-to-Host Bank Disbursement simulated via BIDV Sandbox Adapter (BM02 SR-05)	backend/src/modules/payouts/bidv-payout.adapter.ts	Kept by design due to banking license constraints; BIDV Sandbox idempotency, inquiry reconciliation (SBX-...), FinancePosting, and VietQR batch export verified.	Nguyễn Đình Tuấn	Kept (Sandbox)
G4	Residual deprecated database/mobile stubs after scope removal of standalone promotional portals (BM02 SR-03)	backend/prisma/schema.prisma, Mobile/src/features/workspace/policy.ts	Web routes already redirect/exclude legacy flows and rate parameter is fixed to 0; hide remaining mobile menu entry and mark legacy DB table deprecated in SDD.	Nguyễn Thành Thắng	Defense 1.1

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Multi-Role Auth, 2FA OTP & 5-Role RBAC (Web & Mobile)	All 5 Roles	Argon2id (>=64MB) hashes passwords; >=5 wrong OTPs invalidate code; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden.	TC-F01-01..03	**Done**
F-02	Internal Brand Staff & In-House Store Provisioning	System Administrator	Only SYSTEM_ADMIN creates SHOP_MANAGER (bound to storeId & GHN warehouse) and SYSTEM_MANAGER; public /auth/register with SHOP_MANAGER coerces to CUSTOMER.	TC-F02-01..03	**Done**
F-03	KOC Upgrade eKYC Maker-Checker & Social Channels	Customer, Collaborator, System Manager	Customer submits 12-digit CCCD, bank account & TikTok/FB/YouTube URL; SYSTEM_MANAGER approval sets KycStatus=VERIFIED, role=COLLABORATOR, starting tier=Bronze (10%).	TC-F03-01..03	**Done**
F-04	Immutable Security & Financial Audit Trail	System Administrator, System Manager	100% of staff creations, KYC decisions, MAP updates, VIP deal approvals, return resolutions & payouts append an AuditLog entry; API blocks edit/delete.	TC-F04-01..03	**Done**
F-05	Category-Locked 6-Block Product & Compliance Moderation	Shop Manager, System Manager	Product category locks to store.category (mismatch returns 400); duplicate EAN-13 barcode fails; new SKU stays DRAFT until a System Manager approves (APPROVED).	TC-F05-01..03	**Done**
F-06	MAP Floor-Price Guard & Price Promotion Approval	Shop Manager, System Manager, System Administrator	System Manager sets SKU floorPrice; single/batch promotion with proposedPrice < floorPrice returns 400; discount applies only when status=ACTIVE within budget.	TC-F06-01..03	**Done**
F-07	4-Axis AI Partner Matchmaking & Two-Way Cooperation	Shop Manager, Collaborator	AI scores verified KOCs (0–100 across Category, CR%, Tier/Social, Price Fit) in <=500ms; bilateral partnership invite/request transitions PENDING -> APPROVED.	TC-F07-01..03	**Done**
F-08	Centralized Digital Media Asset Hub	Shop Manager, Collaborator	Shop Manager uploads product-linked HD images/videos/copy; verified Collaborator filters by type and downloads in 1 click; unverified access returns 403.	TC-F08-01..03	**Done**
F-09	Socket.IO Realtime VIP Deal Negotiation (10%–16%)	Shop Manager, Collaborator	Counterparties send/respond to rich VIP Deal cards (targetRevenue, commitmentDays) over /chat WebSocket without reload; 10%–16% deals activate immediately on accept.	TC-F09-01..03	**Done**
F-10	VIP Budget Exception Escalation Gate (17%–25%)	Shop Manager, Collaborator, System Administrator	Proposals >25% fail with 400; accepted 17%–25% deals transition to PENDING_ADMIN_APPROVAL and activate only after SYSTEM_ADMIN approves at /admin/vip-exceptions.	TC-F10-01..03	**Done**
F-11	Sample Product Sponsorship & 14-Day Video Review SLA	Collaborator, Shop Manager, System Manager	Verified KOC requests 0 VND sample; GHN receipt (RECEIVED) starts 14-day SLA timer; unsubmitted video past deadline marks OVERDUE and blocks new sample requests.	TC-F11-01..03	**Done**
F-12	Value-Added Livestream Room & Live Deal Pinning	Collaborator, Shop Manager, Customer	Creator transitions SCHEDULED -> LIVE -> ENDED and pins store SKUs (+2% live bonus); stock=0 disables buy CTA in live tray; ENDED locks pinning and shows summary.	TC-F12-01..03	**Done**
F-13	Smart Link, Dynamic QR & KOC Attribution Coupon	Collaborator, Shop Manager, Customer	Verified KOC creates /r/:shortCode link, QR & goods coupon ([KOC][5|10], shipping vouchers excluded); click sets 30d AttributionSession rate-limited at 100 req/min/IP.	TC-F13-01..03	**Done**
F-14	Authenticated Zero-Loss Checkout & Self-Referral Guard	Customer, Collaborator	Checkout requires login while preserving 100% cart items and settles via PayOS VietQR/Wallet/COD; if buyer matches KOC (isSelfReferral), commission is forced to 0%.	TC-F14-01..03	**Done**
F-15	Multi-Warehouse GHN Fulfillment & Order Tracking	Shop Manager, Customer	Shop Manager confirms order and dispatches GHN parcel from brand warehouse (weight/dimensions); SKU stock decrements atomically and status tracks SHIPPING -> DELIVERED.	TC-F15-01..03	**Done**
F-16	Dual-Path Return Resolution: Exchange vs Refund Clawback	Customer, Shop Manager, System Manager	Within 7-day return window & 48h inspection SLA: EXCHANGE ships 1-1 replacement, preserves 100% KOC commission & opens GHN claim; REFUND credits wallet & reverses commission.	TC-F16-01..03	**Done**
F-17	Best-Benefit Commission, 7-Day Hold & KPI Deal Escrow	Shop Manager, Collaborator, System Manager	Computes min(25%, max(VIP Deal, 10%+Tier, 10%) + Live); holds base fee 7 days in pendingBalance and holds VIP bonus in conditionalDealEscrowBalance until KPI QUALIFIED.	TC-F17-01..03	**Done**
F-18	Monthly Milestone Settlement (FR-09) & Tier Promotion	Shop Manager, Collaborator	Closed-month clean GMV scan excludes refunded/self-referral orders, pays 200k–4M VND milestone bonus idempotently (0 duplicates), and promotes Bronze->Silver->Gold->Diamond.	TC-F18-01..03	**Done**
F-19	Row-Locked Collaborator Withdrawal, 10% PIT & BIDV Payout	Collaborator, System Manager, System Administrator	SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200,000 VND) locks wallet row, withholds flat 10% PIT (FinancePosting), and disburses net via BIDV Sandbox.	TC-F19-01..03	**Done**

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	API & Redirect Latency	REST & Smart Link p95 Response Time	<= 200 ms for catalog queries and /r/:shortCode redirection	Autocannon benchmark (500 concurrent requests) & NestJS timing interceptor
NFR-02	Authentication & RBAC Security	Password Hash Cost & Role Isolation	Argon2id (>= 64 MB memory, 3 iterations); 100% cross-role calls blocked (403)	Jest security suites (auth-otp.spec.ts, rbac guards) across Web & Mobile endpoints
NFR-03	Web & Mobile Client Responsiveness	Web Initial Bundle / FCP & Mobile Bootstrap	Web gzipped entry <= 500 KB, FCP <= 1.5 s; Android /mobile/bootstrap <= 300 ms	Vite build bundle audit, Lighthouse CI, and Mobile contract/startup test suite
NFR-04	Transactional & Inventory Integrity	Atomic Checkout & Return Rollback Rate	100% atomic rollback on stock exhaustion or invalid return state transition	Jest integration specs (orders.service.spec.ts, return-pickup.postgres.integration.spec.ts)
NFR-05	Rate Limiting & Cache Resilience	Click Throttle & Redis Outage Failover	100 req/min/IP limit; 0 unhandled 500 errors during simulated Redis disconnect	CacheService LRU memory fallback unit tests & click-flood load script
NFR-06	Financial Concurrency & Auditability	Double-Spend Prevention & Ledger Coverage	0 negative wallet balances under 20 parallel payouts; 100% mutations in AuditLog	Parallel SELECT ... FOR UPDATE payout concurrency test & FinancePosting assertions

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Nguyễn Thành Thắng (Leader)	Fullstack Architect, IAM & Brand Governance Lead	5-Role IAM & Staff Provisioning (F-01..F-04); Category-Locked Catalog, MAP Pricing, AI Matchmaking & Media Hub (F-05..F-08)	WF-01: Internal Staff & KOC eKYC Onboarding; WF-02: Brand Product Compliance, MAP Pricing & AI Matchmaking	Decisions D1 & D2; Auth, Users, KYC, Products, Pricing & AI Matchmaking modules
Phan Xuân Thịnh	Fullstack & Mobile/Realtime Lead	Socket.IO VIP Deal Negotiation & >16% Admin Gate (F-09, F-10); 14-Day Sample SLA (F-11); Live Pinning & Mobile Client (F-12)	WF-03: Socket.IO VIP Deal Negotiation, 14-Day Sample Sponsorship SLA & Live Deal Pinning	Decision D3; Chat, AffiliateDeals, Samples, LiveSessions & Expo Mobile modules
Nguyễn Phú Quý	Fullstack, Attribution & Reverse Logistics Lead	Smart Link/QR & KOC Coupon Attribution (F-13); Auth Checkout & Self-Referral Guard (F-14); GHN & Dual-Path Returns (F-15, F-16)	WF-04: Multi-Channel KOC Attribution, Anti-Self-Referral Checkout, GHN Fulfillment & Dual-Path Returns	Decision D4; ReferralLinks, Checkout, Orders, Shipping (GHN) & Returns modules
Nguyễn Đình Tuấn	Backend & Financial Settlement Lead	Best-Benefit Commission, 7-Day Hold & KPI Escrow (F-17); Monthly Milestone FR-09 & Tiers (F-18); 10% PIT & BIDV Payout (F-19)	WF-05: 7-Day Escrow Maturation, Monthly Milestone Settlement (FR-09), 10% PIT Tax & BIDV Sandbox Payout	Decision D5; Commissions, CommissionRules, Wallets, FinancePosting & Payouts modules

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Internal Brand Staff Provisioning, KOC eKYC Onboarding & Social Verification	System Administrator, Customer, System Manager, Collaborator	System Admin appoints an internal Brand Manager (/admin/personnel), or a Customer registers, verifies 2FA OTP, and submits a KOC upgrade eKYC dossier (/customer/upgrade).	In-house Shop Manager bound to assigned brand store; verified Customer upgraded to Collaborator at Bronze (10%) tier with verified social channel and immutable AuditLog.	F-01, F-02, F-03, F-04	User: isActive=true (SHOP_MANAGER/COLLABORATOR); KycStatus: UNVERIFIED -> VERIFIED / REJECTED; Tier: Bronze (10%)	Nguyễn Thành Thắng
WF-02	Category-Locked Product Compliance, MAP Floor-Price Promotion, AI Matchmaking & Media Hub	Shop Manager, System Manager, Collaborator	Shop Manager creates a 6-block product SKU, proposes a price drop, uploads Media Hub assets, and runs 4-Axis AI Partner Matchmaking.	Category-locked SKU moderated to APPROVED, MAP-compliant price promotion activated, Top KOC matched & approved for store cooperation, and HD media downloaded.	F-05, F-06, F-07, F-08	ProductModerationStatus: DRAFT -> APPROVED / REJECTED; PricePromotionStatus: PENDING -> APPROVED -> ACTIVE; StoreCollaboratorStatus: PENDING -> APPROVED	Nguyễn Thành Thắng
WF-03	Socket.IO VIP Deal Negotiation (>16% Admin Gate), 14-Day Sample SLA & Live Pinning	Shop Manager, Collaborator, System Administrator, Customer	Approved Shop Manager and Collaborator negotiate an Exclusive VIP Deal card in Socket.IO Chat, dispatch a 0 VND product sample, and host a live product-pinning session.	VIP Deal activated (immediately for <=16% or after System Admin approval for 17%–25%), sample video verified within 14-day SLA, and live pinned SKU earns +2% live bonus.	F-09, F-10, F-11, F-12	ExclusiveDealStatus: PENDING -> PENDING_ADMIN_APPROVAL -> APPROVED; SampleRequestStatus: PENDING -> APPROVED -> SHIPPED -> RECEIVED -> VIDEO_SUBMITTED -> COMPLETED / OVERDUE; LiveSessionStatus: SCHEDULED -> LIVE -> ENDED	Phan Xuân Thịnh
WF-04	Multi-Channel KOC Attribution, Self-Referral Guard Checkout, GHN & Dual-Path Returns	Collaborator, Customer, Shop Manager, System Manager	Collaborator shares Smart Link/QR and KOC Attribution Coupon; Customer checks out, receives GHN parcel, and files either a damaged-item Exchange or a Refund return.	Order attributed accurately (0% if self-referral), shipped via brand GHN warehouse; 1-for-1 EXCHANGE preserves 100% KOC commission (+ GHN claim) while REFUND claws back commission.	F-13, F-14, F-15, F-16	OrderStatus: PENDING -> SHIPPING -> DELIVERED -> COMPLETED / RETURNED; ReturnRequestStatus: REQUESTED -> SHOP_APPROVED -> INSPECTING -> EXCHANGE_SHIPPED / REFUNDED -> COMPLETED; CommissionStatus: PENDING -> APPROVED / REVERSED	Nguyễn Phú Quý
WF-05	7-Day Escrow Maturation, Monthly Milestone Settlement (FR-09), 10% PIT & BIDV Payout	Shop Manager, Collaborator, System Manager, System Administrator	7-day post-delivery hold elapses on eligible orders, Shop Manager settles closed-month FR-09 milestones, and Collaborator submits a bank withdrawal request.	Mature commissions and qualified KPI bonuses unlock to availableBalance, monthly bonus & tier upgrade apply once, 10% flat PIT is withheld under DB row lock, and net payout disburses via BIDV Sandbox.	F-17, F-18, F-19	CommissionStatus: PENDING -> APPROVED; DealBonusStatus: PENDING_KPI -> QUALIFIED / FORFEITED; PayoutStatus: PENDING -> PROCESSING -> APPROVED / REJECTED	Nguyễn Đình Tuấn

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	WF-01	**Main**	Seeded System Admin, System Manager, and new Customer account; Sora Skincare store configured.	1. Admin creates internal SHOP_MANAGER bound to Sora Skincare at /admin/personnel. 2. Customer registers, verifies OTP, and submits CCCD, MB Bank & TikTok channel at /customer/upgrade. 3. System Manager approves eKYC on Web/Mobile.	Staff account active on Sora Skincare; Customer upgraded to COLLABORATOR with KycStatus=VERIFIED, initial tier Bronze (10%), and AuditLog recorded.	Nguyễn Thành Thắng

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01b	WF-01	**Exception**	Unauthenticated client sends role=SHOP_MANAGER to /auth/register; KOC submits blurred CCCD or >5 wrong OTPs.	1. Call POST /auth/register with role=SHOP_MANAGER. 2. Enter wrong OTP 5 times. 3. System Manager rejects mismatched KYC bank/CCCD dossier with mandatory reason.	Public signup coerced to CUSTOMER (privilege escalation blocked); OTP invalidated after 5 failures; KycStatus=REJECTED with feedback and resubmission enabled.	Nguyễn Thành Thắng
DS-02a	WF-02	**Main**	Sora Skincare (locked to Cosmetics), verified KOC profiles, and SKU floorPrice = 350,000 VND.	1. Shop Manager creates Vitamin C Serum (450k VND, EAN-13); System Manager approves. 2. Shop Manager proposes 390k VND promotion (>= floorPrice); System Manager approves. 3. Run 4-Axis AI Matchmaker, approve KOC cooperation, and KOC downloads HD media.	Product APPROVED on storefront; promotion ACTIVE at 390k VND; AI returns ranked KOC radar scores; StoreCollaborator=APPROVED and HD asset downloaded.	Nguyễn Thành Thắng
DS-02b	WF-02	**Exception**	Shop Manager attempts cross-category SKU, duplicate EAN-13 barcode, or price drop below floorPrice (320k < 350k VND).	1. Submit Electronics category from Cosmetics store. 2. Submit duplicate EAN-13 barcode. 3. Propose promotion price 320,000 VND below 350,000 VND MAP floor.	All three requests rejected with 400 Bad Request; product and price remain unchanged, protecting brand vertical isolation and unit margin.	Nguyễn Thành Thắng
DS-03a	WF-03	**Main**	Approved Sora–KOC partnership on Web/Mobile; approved Vitamin C Serum with sample quota > 0.	1. Shop sends 16% VIP Deal card (KPI 30M VND) via Socket.IO Chat; KOC accepts (auto-APPROVED); also test 18% deal escalating to Admin approval. 2. KOC requests sample -> Shop dispatches GHN -> KOC submits review URL within 14d -> Shop marks COMPLETED. 3. KOC pins SKU in LIVE room.	16% VIP deal active immediately (18% active after Admin approval); sample reaches COMPLETED within 14-day SLA; LIVE tray pins SKU with +2% bonus.	Phan Xuân Thịnh
DS-03b	WF-03	**Exception**	VIP Deal proposed at 28% (>25% cap); KOC misses 14-day sample review deadline; pinned SKU stock hits 0.	1. Propose 28% VIP Deal in chat. 2. Trigger overdue scan on sample past 14-day deadline and request another sample. 3. Set pinned SKU stock = 0 during LIVE session.	28% deal blocked (400 Bad Request); sample transitions to OVERDUE and blocks new sample requests; out-of-stock SKU disables purchase CTA in live tray.	Phan Xuân Thịnh
DS-04a	WF-04	**Main**	Verified KOC with active 16% VIP deal & KOC coupon (10% off); authenticated Customer; Sora GHN warehouse.	1. KOC generates /r/:shortCode, QR & KOC coupon. 2. Customer clicks link / applies KOC coupon, logs in preserving 100% cart, pays via PayOS/COD. 3. Shop creates GHN waybill -> DELIVERED. 4. On transit damage, Customer chooses EXCHANGE -> Shop ships 1-1 replacement & opens GHN claim.	Order attributed to KOC with pending commission recorded; replacement delivered (COMPLETED), 100% KOC commission preserved, and GHN compensation claim opened.	Nguyễn Phú Quý
DS-04b	WF-04	**Exception**	KOC attempts self-purchase via own link/coupon; Customer requests full return & refund within 7-day window.	1. KOC checks out using own referral link/matching phone. 2. Separate Customer files REFUND return with unboxing video -> parcel returned & inspected -> refund approved.	Self-referral order sets commission rate = 0% and flags AiFraudService; refunded order credits Customer shoppingBalance and reverses KOC commission to REVERSED.	Nguyễn Phú Quý
DS-05a	WF-05	**Main**	Delivered order past 7-day hold with no open return; closed-month clean GMV > 50M VND; verified KOC wallet.	1. Reconciliation job matures 7-day PENDING commission to availableBalance and qualifies KPI deal escrow. 2. Shop settles closed-month FR-09 milestone bonus & promotes KOC tier. 3. KOC requests 2,072,000 VND withdrawal -> System Manager approves via BIDV Sandbox.	Wallet debited once under SELECT ... FOR UPDATE; 207,200 VND (10% PIT) posted via FinancePosting; 1,864,800 VND net approved with SBX-... reference and AuditLog.	Nguyễn Đình Tuấn
DS-05b	WF-05	**Exception**	SHOP_MANAGER attempts withdrawal; KOC requests < 200,000 VND or fires parallel requests exceeding balance; BIDV Sandbox timeout.	1. Call POST /payouts/withdraw as SHOP_MANAGER. 2. Submit 100,000 VND (<200k min) and 2 concurrent full-balance requests. 3. Simulate BIDV timeout/rejection.	SHOP_MANAGER blocked (403 Forbidden); <200k rejected (400); row lock allows at most 1 request (0 negative balance); rejected payout restores 100% funds via PAYOUT_REJECT_REFUND.	Nguyễn Đình Tuấn

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features (from Section 6)	19	3	57
Main workflows (end-to-end, from A1)	5	1	5
State transitions of main entities (KYC, Product, PricePromo, Deal, Sample, Live, Order, Return, Commission, Payout)	13	1	13
Agreed NFRs & concurrency/security suites (from Section 7)	6	1	6
Committed total (minimum)	—	—	**81**